

Timeformer: Explicit Temporal Conditioning for Traceable Semantic Representations

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Abstract. Words change meaning over time, yet most computational models treat token representations as epoch-independent. We study temporal traceability: the property that the same surface token, queried at different epochs, occupies semantically appropriate neighborhoods in representation space. A model may use epoch information to improve predictive accuracy without exposing a traceable `token@time` representation: this gap is the focus of the paper.

We address it with Timeformer, a family of Transformer-style encoders organized as a controlled ablation chain. Four variants add temporal capacity in successive steps: a static baseline, a globally time-conditioned encoder, a token-time encoder that conditions each token embedding on the epoch before self-attention is applied, and a memory-augmented extension that consults causal historical prototypes. All variants are trained with masked language modeling on a controlled synthetic benchmark in which the temporal trajectory of each target item is known and evaluation labels are withheld from training.

The main finding is that explicit token-time interaction is the effective mechanism for traceability. The Token-Time Transformer produces a neighborhood drift of -0.614 for drifting subjects across epochs, compared to -0.231 for the static baseline (a difference stable across seeds and visible in class-specific trajectory curves). Probe-based accuracy gains are modest and seed-sensitive, confirming that geometric diagnostics are more appropriate than label recoverability for evaluating traceability. The memory-augmented variant does not produce a robust improvement over token-time interaction; the oracle diagnostic identifies mean-prototype aggregation as the failure point, motivating sense-aware memory mechanisms in future work.

Keywords: Temporal semantics · Semantic change · Transformer models · Traceability · Diagnostic evaluation

1 Introduction

Words do not preserve fixed meanings across time. In 1900, *broadcast* referred to scattering seeds across a field; by mid-century it denoted radio and television transmission; today it describes a social media post reaching millions of users. Most language models, however, take no account of when a text was written: a word’s representation depends on its surrounding context, but not

on the historical period in which it appeared. A model that cannot distinguish `broadcast@1900` from `broadcast@2020` cannot be inspected for semantic movement across time.

The ability to query a word at a specific epoch matters beyond technical benchmarks. Consider *gay*, which in the early twentieth century was predominantly associated with terms such as *cheerful*, *merry*, and *carefree*, and by the late twentieth century with *queer*, *lesbian*, and *bisexual* [3]. Whether this transition involved a shift toward or away from terms linked to social stigma, and when exactly that shift occurred, is an empirical question. To answer it, one needs a representation that exposes `gay@1920` and `gay@1980` as directly comparable objects in a shared space. Without such a representation, the question can only be addressed through aggregate statistics that conflate meaning with unrelated features of the corpus.

The dominant computational approach, lexical semantic change detection (LSCD), measures this kind of shift by training distributional representations on time-sliced corpora and comparing them across periods [6, 7]. One model may associate *broadcast* in 1900 with *sow* and *field*, while another, trained on 2020 data, associates it with *signal* and *audience*; semantic change is inferred from the distance between the two spaces. This approach has produced important methods and benchmarks, but it summarizes the change as a single similarity score between two separately trained models, not as a queryable `token@time` object. `broadcast@1900` and `broadcast@2020` are implicit in the models but not directly inspectable within a single shared space.

A subtler limitation arises when a single model receives temporal information during training. A model may correctly interpret *virus* as biological infection in a 1900 sentence and as malicious software in a 2020 sentence, improving its word-prediction accuracy, while still assigning nearly identical neighborhoods to the token across epochs. Temporal information can improve prediction without producing a representation in which the same word occupies measurably different neighborhoods at different epochs. Prediction accuracy and geometric traceability are separable properties.

We call a representation *temporally traceable* if the same surface token, queried at different epochs, occupies neighborhoods consistent with its meaning at that epoch, making `token@time` a directly inspectable object. This paper introduces Timeformer, a family of Transformer-style encoders designed around this property. The family is organized as a controlled ablation chain: a Static Transformer with no time information; an Additive Time-Conditioned Transformer that injects a global epoch vector; a Token-Time Transformer that conditions each token embedding on the epoch before self-attention is applied; and a Memory-Augmented Timeformer that consults causal historical prototypes. This design lets us isolate which form of temporal conditioning produces traceability, rather than treating it as a single architectural choice. We study this in a controlled synthetic benchmark in which the temporal trajectory of each subject is known, giving diagnostic access to target neighborhoods without the confounds of natural historical corpora.

The paper makes four contributions: it formulates temporal traceability as a `token@time` representation problem; introduces the Timeformer ablation family; evaluates traceability with diagnostics that inspect subject representations directly (context drift score, contrastive sign-flip rate, and probe-based evaluation); and shows that token-time interaction is the main positive mechanism, while mean-pooled historical memory is better understood as a diagnostic extension. Section 2 reviews related work; Sections 3–5 describe the benchmark, architecture, and evaluation protocol; Section 6 reports results; Sections 7–8 discuss and conclude.

2 Related Work

Computational approaches to semantic change are commonly framed as lexical semantic change detection: given language data from different periods, the goal is to determine whether and how the meaning of a target item has changed. Diachronic word embeddings provided one of the most influential formulations of this problem by learning distributional representations for different time slices and comparing them across periods [3]. This family of methods made semantic change operational through geometric movement in embedding space, often after alignment or normalization across independently trained temporal slices. The central intuition remains important for this paper: semantic change should be visible as a change in neighborhood structure. Our focus, however, is not on aligning independently trained spaces. Instead, we ask whether a single Transformer-style encoder can expose a representation that is directly queryable as a lexical item at a specific epoch.

Benchmarks for LSCD have helped standardize evaluation by defining target words, periods, and gold annotations for semantic change [6, 7]. These resources are essential for evaluating change in natural corpora, but they inherit the observational complexity of historical language, where changes in frequency, genre, and corpus composition interact with lexical meaning. We use a controlled benchmark as a diagnostic testbed rather than a replacement for natural LSCD benchmarks, trading observational realism for direct access to planted trajectories and evaluation labels.

A second relevant line of work concerns temporal conditioning in neural models. Rather than training separate models for each period, a temporally conditioned model can receive time as part of its input or architecture, allowing parameter sharing across epochs while representations or predictions vary with time [1, 5, 4]. For the traceability problem, however, the form of conditioning matters. A global epoch vector may tell the model which period it is processing, but it does not necessarily force an interaction between a particular token identity and that epoch. Timeformer is therefore organized as an ablation chain that separates global time injection from explicit token-time interaction. This distinction is important because the paper evaluates not only whether time improves a prediction, but whether the same token occupies different and interpretable neighborhoods at different epochs.

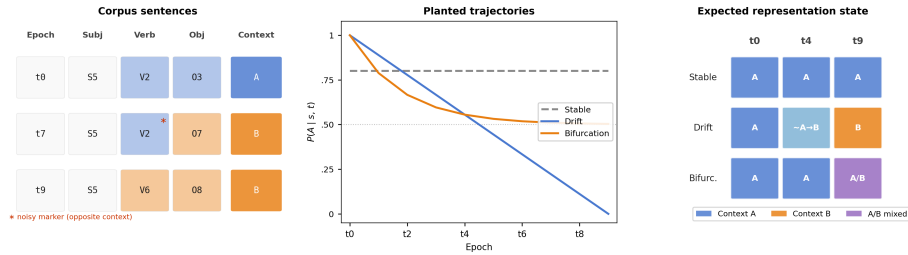


Fig. 1. Benchmark design overview. *Left:* SVO sentences for a drifting subject across three epochs; an N_1 verb appearing in an N_2 sentence at $t7$ (marked *) illustrates the deliberate noise in local markers. *Center:* planted $P(N_1 | s, t)$ trajectories for Stable (flat), Drift (monotonic), and Bifurcating (levelling at 0.5) subjects across $t0$ – $t9$. *Right:* expected nearest-neighbor behavior for a traceable representation.

Finally, this paper draws on diagnostic evaluation and multi-sense representations. Probe accuracy does not establish traceability, so we combine it with geometric diagnostics that inspect the subject token directly. Semantic change may involve bifurcation rather than a simple shift [2]; the memory-augmented variant exposes why mean-pooled prototypes are insufficient for that structure.

3 Controlled Temporal Benchmark

The benchmark gives each subject a known temporal trajectory, making it possible to ask directly whether a model’s representation of a subject at $t0$ has moved toward the expected neighborhood by $t9$. This distinguishes it from natural-language LSCD benchmarks, where ground truth is often annotated or inferred post-hoc and corpus confounds are hard to separate from genuine semantic change.

The corpus uses a subject–verb–object (SVO) grammar: every sentence consists of one subject, one verb, and one object. Two semantic neighborhoods, N_1 and N_2 , partition the vocabulary of 8 verbs and 8 objects: N_1 uses $\{V1, \dots, V4\}$ and $\{O1, \dots, O4\}$; N_2 uses $\{V5, \dots, V8\}$ and $\{O5, \dots, O8\}$. A sentence generated from N_1 therefore looks like S5 V2 O3, while one from N_2 looks like S5 V6 O8; the subject token is the same in both cases, but its co-occurring words differ. Intuitively, the intended meaning of subject s at epoch t is how often it appears in N_1 sentences; formally, $P(N_1 | s, t)$ is the probability that a sentence is drawn from N_1 . The vocabulary contains 30 subjects; each subject’s trajectory is determined by how $P(N_1 | s, t)$ evolves across the 10 epochs $t0$ through $t9$.

Figure 1 (center) shows the three planted trajectory classes. Stable subjects maintain an approximately constant $P(N_1 | s, t)$ throughout. Drift subjects follow a monotonic trajectory from near-exclusive N_1 ($P(N_1 | s, t) \approx 1$ at $t0$) to near-exclusive N_2 ($P(N_1 | s, t) \approx 0$ at $t9$): a drifting subject that appears in sentences like S5 V2 O3 early on will appear in sentences like S5 V6 O8 by the final epoch. Bifurcating subjects begin in N_1 and move toward a mixed state

Table 1. Model variants in the Timeformer ablation family.

ID	Public name	Temporal signal	Memory
Static	Static Transformer	None	No
Additive	Additive Time- Conditioned Transformer	Global epoch vector	No
Joint	Token-Time Transformer	Token-time interaction	No
Timeformer	Memory-Augmented Timeformer	Token-time interaction	Causal proto- types

($P(N_1 \mid s, t) \approx 0.5$) where both neighborhoods appear with equal frequency, approximating two coexisting senses. The benchmark contains 10 subjects per class.

At epoch $t7$, for example, a sentence for **S5** might read **S5 V2 07** (Figure 1, left): the planted neighborhood is N_2 , but **V2** belongs to N_1 . Each local marker is drawn from the opposite neighborhood 25% of the time, so the model cannot recover the planted neighborhood from the words alone. It must use subject identity together with epoch. The `true_context` label is determined by the planted trajectory and is withheld from training; it is used only for evaluation.

The benchmark supports several evaluation conditions. The standard split uses the same 75% marker accuracy as training. Two harder splits force the verb, or both verb and object, to come from the opposite context. An ambiguous split sets fidelity to 50%, so local markers carry no reliable signal; only subject identity and epoch remain. Two additional conditions test temporal behaviour directly and are described in Section 5: the contrastive set presents the same sentence under two different epoch labels to check whether changing only the epoch changes the preferred context; the continuation set uses epochs held out from training to test generalisation. Results throughout are diagnostic evidence in a controlled setting, not a claim about open-domain historical language.

4 Timeformer Architecture

Timeformer is organized as a controlled ablation chain: each variant adds exactly one new capability over the previous one, so that any difference in traceability can be attributed to that single change. Figure 2 shows the complete forward path; the simpler variants are special cases of the same diagram. All variants share the same training interface: a sentence from the SVO corpus (Section 3) is encoded as `[CLS] S V O [SEP]`, a Transformer encoder produces contextual hidden states, and h_s – the final hidden state at the subject position – is the representation used for all traceability diagnostics. Variants differ only in which boxes in Figure 2 are active. Table 1 summarizes the four variants.

4.1 Static Transformer

The Static Transformer (Static) uses only the Token input box in Figure 2: no Time input, no Token-Time Interaction. For each input token w_i , the embedding

Figure 2. Timeformer ablation chain

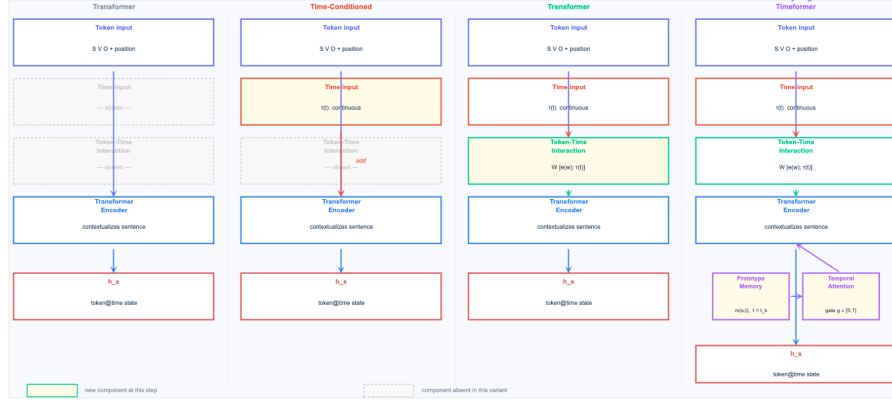


Fig. 2. Timeformer ablation chain. Each column adds exactly one new component (highlighted in yellow) over the previous variant. **Static** (Static Transformer) encodes tokens with no temporal information. **Additive** introduces the Time input $\tau(t)$ added directly to token embeddings. **Joint** (Token-Time) routes both token and time through a joint projection $W[e(w); \tau(t)]$ via the Token-Time Interaction box. **Timeformer** (Memory-Augmented) extends Joint with Prototype Memory and Temporal Attention that gate historical context into the encoder. Dashed grey boxes mark components absent in a given variant.

combines a learned token embedding $e(w_i) \in \mathbb{R}^d$ with a fixed sinusoidal position embedding $p_i \in \mathbb{R}^d$:

$$x_i = e(w_i) + p_i.$$

The sequence is passed through a pre-layer-normalized Transformer encoder with no epoch signal. As a consequence, the sentence S5 V2 O3 at t_0 and the same sentence at t_9 produce the same h_s : without a temporal signal, the model cannot distinguish them. This is the null hypothesis for traceability.

4.2 Shared Component: Continuous Time Encoding

The Time input box (lower left of Figure 2) is shared by Additive, Joint, and Timeformer. Rather than a learned lookup table – which would memorize training epochs as unrelated identifiers – a continuous encoding treats epochs as ordered points on a shared scale, supporting interpolation and extrapolation. The epoch index t is normalized to $[0, 1]$, passed through fixed sinusoidal features, and projected by a small MLP:

$$\tau(t) = \text{MLP}(\text{sinusoidal}(t/T)).$$

These features encode epoch index only and are entirely distinct from the token-position encodings already present in the Transformer.

4.3 Additive Time-Conditioned Transformer

The Additive variant routes $\tau(t)$ from the Time input box directly into the token embeddings by addition, without passing through the Token-Time Interaction box:

$$x_i = e(w_i) + p_i + \tau(t).$$

Because the same $\tau(t)$ is added to every token regardless of its identity, the subject, verb, and object all receive the same temporal push. The model learns that a sentence belongs to epoch t , but not that subject S5 specifically has shifted meaning by t . This variant isolates whether global period information alone is sufficient for traceability.

4.4 Token-Time Transformer

The Token-Time Transformer (Joint) activates the Token-Time Interaction box in Figure 2: the token embedding and time encoding are concatenated and jointly projected before self-attention:

$$x_i = W[e(w_i); \tau(t)] + p_i,$$

where $e(w_i), \tau(t) \in \mathbb{R}^d$ and $W \in \mathbb{R}^{d \times 2d}$. The projection W sees token identity and epoch simultaneously, so the same epoch can affect different tokens differently. Concretely, S5 at t_0 receives $W[e(\text{S5}); \tau(0)] + p_1$ while S5 at t_9 receives $W[e(\text{S5}); \tau(9)] + p_1$: the initial embedding differs before any self-attention, giving the subject a distinct starting point at each epoch.

4.5 Memory-Augmented Timeformer

The Memory-Augmented Timeformer adds the lower section of Figure 2: a Prototype Memory and Temporal Attention gate that update the subject embedding before it enters the encoder. For each subject s and epoch t , prototype $m(s, t)$ is the mean of s 's hidden states across all training sentences at that epoch. Prototypes are fixed during the forward pass (no gradients flow back through them) and served in causal order: at t_k , the model attends only to $\{m(s, t) : t < t_k\}$. For example, at t_7 the memory for S5 is $\{m(\text{S5}, 0), \dots, m(\text{S5}, 6)\}$ – a summary of how that subject has appeared in each of the six preceding epochs.

The cross-attention layer uses the current subject embedding as the query and the historical prototypes as keys and values. The updated embedding is:

$$\tilde{x}_s = x_s + g \cdot (\text{LN}(x_s + \text{Attn}(x_s, M_s, M_s)) - x_s),$$

where M_s collects the causal prototypes for subject s . The scalar gate $g \in [0, 1]$ is initialized to zero: when $g = 0$ the output equals x_s unchanged; when $g = 1$ it is the layer-normalized result of attending over past prototypes. The model opens the memory channel only if it proves useful under the MLM objective.

5 Evaluation Protocol

All variants are trained with masked language modeling over the synthetic SVO corpus. The planted `true_context` label is never used during training; it is used only after training to evaluate whether the learned representation contains temporal semantic information. This separation reflects the intended use case of representation learning from corpora without semantic-change labels. The primary evaluation target is h_s , the final hidden state at the subject position, since subjects are the lexical items with planted temporal trajectories.

We use four complementary diagnostics. First, a linear probe is trained on h_s from the standard test split to predict `true_context` and is then evaluated on each split. This measures whether context information is linearly recoverable from the representation, but it does not by itself establish traceability. Second, we compute a context drift score over nearest neighbors. For each epoch and subject class, we retrieve the k nearest neighbors of each subject representation and measure the proportion assigned to N_1 , with k fixed at the same value across all models and splits. A traceable representation should show a strong movement from N_1 -like to N_2 -like neighborhoods for drifting subjects, weaker movement for bifurcating subjects, and comparatively limited movement for stable subjects. Third, we use contrastive temporal examples in which the same subject token is presented with two different epoch labels. We define the contrastive sign-flip rate as the proportion of such pairs for which swapping the epoch label reverses the nearest-neighbor assignment (from N_1 -dominant to N_2 -dominant or vice versa). A model without temporal capacity is expected to produce a sign-flip rate of zero, since the same token yields the same representation regardless of the epoch label provided. Fourth, we evaluate continuation examples from held-out later epochs to test whether temporal behavior extends beyond the training distribution.

Comparisons follow the ablation chain: Additive vs. Static isolates the effect of global time conditioning; Joint vs. Additive isolates token-time interaction; Timeformer vs. Joint isolates the memory contribution, evaluated on continuation and contrastive diagnostics.

For the memory-augmented model, we include a shuffled-memory control (a subject receives prototypes from a different subject), a no-history control (the memory path is present but empty), and an oracle-memory diagnostic (the model receives perfect historical prototypes). Memory is considered useful only when it improves over the Token-Time Transformer and over both inappropriate and empty histories.

6 Results

We report results over three training runs. Unless otherwise noted, values are means over seeds, with the standard deviation shown in parentheses when space allows. The main finding is that token-time interaction produces a much clearer temporal movement in representation space than the Static Transformer. The memory-augmented variant is informative as a diagnostic extension, but it does not provide a robust improvement over the Token-Time Transformer.



Fig. 3. Context drift score across epochs. The Token-Time Transformer shows stronger movement from N_1 toward N_2 for drifting subjects than the Static Transformer, while stable and bifurcating subjects provide class-specific controls.

Table 2. Context drift score at the first and last epochs. Values are N_1 proportions among nearest neighbors; $\Delta = t_9 - t_0$.

Class	Model	t0	t9	Δ
Stable	Static Transformer	0.736	0.756	+0.020
Stable	Token-Time Transformer	0.843	0.641	-0.203
Stable	Memory-Augmented Timeformer	0.810	0.658	-0.153
Drift	Static Transformer	0.618	0.387	-0.231
Drift	Token-Time Transformer	0.847	0.233	-0.614
Drift	Memory-Augmented Timeformer	0.836	0.263	-0.573
Bifurcation	Static Transformer	0.794	0.682	-0.112
Bifurcation	Token-Time Transformer	0.895	0.521	-0.374
Bifurcation	Memory-Augmented Timeformer	0.905	0.517	-0.388

6.1 Temporal Neighborhood Movement

Figure 3 and Table 2 show the context drift score across epochs. For drifting subjects, the Static Transformer drops from 0.618 to 0.387 ($\Delta = -0.231$), while the Token-Time Transformer shows a much stronger transition from 0.847 to 0.233 ($\Delta = -0.614$). This is the clearest evidence for temporal traceability: the same subject token occupies substantially different neighborhoods when queried at different epochs.

The class-specific curves serve as controls (Table 2). Stable subjects show much smaller movement and no coherent A-to-B trend. Bifurcating subjects show intermediate drift consistent with a trajectory toward a mixed state, not a complete transition. The geometric diagnostic thereby distinguishes all three trajectory classes without relying on a single aggregate score.

Table 3. Main traceability diagnostics. Accuracy columns report linear-probe accuracy on h_s ; flip is contrastive sign-flip rate. A dash indicates the metric was not computed for this variant.

Model	Drift	Flip	Ambig.	Cont.
Static Transformer	-0.231	0.000	0.577	0.719
Additive Time-Conditioned Transformer	—	0.690	0.573	0.754
Token-Time Transformer	-0.614	0.517	0.599	0.766
Memory-Augmented Time-former	-0.573	0.506	0.599	0.766

6.2 Probe and Contrastive Diagnostics

The linear probe results are supportive but less decisive than the neighborhood analysis. On the ambiguous split, where local verb and object markers are uninformative, the Static Transformer reaches 0.577 accuracy, the Additive Time-Conditioned Transformer reaches 0.573, and the Token-Time Transformer reaches 0.599. The corresponding mean delta for token-time interaction is +0.026, but it is seed-sensitive: the three runs give +0.053, +0.005, and +0.021. This confirms that probe accuracy alone should not carry the central claim. It measures recoverability of the planted label, not the geometry of temporal movement.

The contrastive sign-flip diagnostic gives a different view. The Static Transformer has a sign-flip rate of 0.000, as expected. The Additive variant averages 0.690 while the Token-Time Transformer averages 0.517, an ordering that may appear counterintuitive. The decoupling arises because sign-flip counts threshold crossings in individual contrastive pairs, whereas drift score measures the full trajectory magnitude. A global additive offset displaces all tokens uniformly, generating many crossings without coherent subject-level movement. Token-time interaction produces more structured, subject-specific movement (evidenced by the stronger drift) but does not necessarily cross the threshold in more pairs. We therefore treat sign flip as evidence that time conditioning changes behavior at all, and the neighborhood trajectories as the more diagnostic measure of traceability.

6.3 Memory Diagnostics

The Memory-Augmented Timeformer does not produce a robust gain over the Token-Time Transformer. On continuation, both models average 0.766, and the per-seed memory deltas are +0.013, +0.005, and −0.018. The oracle diagnostic clarifies the failure mode: oracle memory improves contrastive sign flip (0.586 vs. 0.448 for the Token-Time reference) but not continuation (0.754 vs. 0.762). Perfect historical prototypes can affect contrastive temporal behavior, but do not solve the continuation objective under the current MLM setup, supporting the use of Timeformer as a diagnostic extension.

Table 4. Memory diagnostics from the corrected oracle run. Continuation is linear-probe accuracy on held-out later epochs; flip is contrastive sign-flip rate.

Variant	Continuation	Sign flip	Delta vs. Joint ref.
Joint reference	0.762	0.448	–
Timeformer learned	0.775	0.448	+0.013 cont.; +0.000 flip
Timeformer oracle	0.754	0.586	-0.008 cont.; +0.138 flip

7 Discussion

The results support a clear conclusion: temporal traceability is a geometric property that does not reduce to predictive accuracy. The Token-Time Transformer improves probe accuracy by only +0.026 on average (modest and seed-sensitive), yet shows a neighborhood drift of -0.614 vs. -0.231 for the static baseline. This dissociation is the central empirical finding: a model can use epoch information to improve predictions without exposing a representation in which the same lexical item occupies meaningfully different neighborhoods at different epochs. Evaluating traceability therefore requires geometric diagnostics, not only label recoverability.

The additive variant shows that knowing the epoch is not sufficient. A global additive offset shifts all token representations by the same vector, leaving subject-level trajectory geometry weakly constrained. Token-time interaction changes this: concatenating each token embedding with the epoch encoding before projection allows the same epoch to affect different tokens differently, so the same subject receives a distinct embedding at each epoch. This is the mechanism behind the coherent neighborhood trajectories in the drift analysis.

The Memory-Augmented Timeformer tests the simplest form of causal memory: a mean of historical subject representations per epoch. Learned prototypes produce near-zero average gain over the Token-Time Transformer, and the oracle diagnostic identifies why: perfect historical prototypes improve contrastive sign-flip rate but not continuation accuracy. For bifurcating subjects, which develop two coexisting senses in later epochs, the mean prototype is a centroid between two clusters, representing neither sense well. This motivates sense-aware memory mechanisms that preserve the internal structure of historical representations rather than collapsing it, and constitutes the main open direction from this work.

8 Conclusion

This paper studied whether a Transformer-style encoder can expose a temporally traceable representation, meaning one in which the same lexical item, queried at different epochs, occupies neighborhoods that reflect its intended temporal context. The answer, in the controlled benchmark used here, is conditional: explicit token-time interaction is necessary. A static encoder and a globally time-conditioned encoder both fail to produce coherent subject-level trajectories. An

encoder that conditions each token embedding on the epoch before self-attention is applied does produce such trajectories, with a neighborhood drift for drifting subjects that is more than twice that of the static baseline (-0.614 vs. -0.231).

The contribution is narrow and deliberate: token-time interaction supports traceability in a controlled diagnostic setting, not a claim about open-domain historical language. The evaluation protocol also contributes a method for separating representational traceability from predictive performance. Mean-pooled historical prototypes do not produce a robust improvement, and the oracle diagnostic identifies single-prototype aggregation as the failure point. Sense-aware memory mechanisms are the natural next step.

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